

Love of Variety and Strategic Advertising: Evidence from Scanner Data

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Abstract

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1 Introduction

In the baseline logit model, a consumer is perceived as having a fixed preference for each product, and the utility from consuming a product is independent of past consumption. Further, it is assumed that a new good is always a utility boosting event, by the error term's property of being identically and independently distributed. However, in many markets, consumers may have a love of variety, or a desire to consume different products over time. This can be due to factors such as boredom, satiation, or a preference for novelty. In such cases, the utility from consuming a product may depend on past consumption, and the introduction of a new good may not always lead to an increase in utility.

This paper develops a dynamic discrete choice model of consumer demand that incorporates a love of variety, and estimates the model using scanner data. The model allows for both forward-looking and myopic consumers, and includes a state variable that captures the mean consumption of each product. The model also incorporates strategic advertising by firms, which can influence consumer preferences and demand.

This idea is motivated by the idea that there are markets which see firms both acting as the host of the market as well as a participant in it. For example, in the grocery market, a firm like Kroger is both a retailer that hosts other firms' products, as well as a producer of its own goods. This may present a market power dynamic where the player with a dual role may have more access to consumer histories and preferences, allowing for more effective advertising via targeting. This paper seeks to understand the implications of this dynamic for consumer welfare and market outcomes. This will be accomplished by estimating the model using scanner data, and conducting counterfactual analyses to evaluate the effects of different market structures.

The primary contribution of this paper is wedding the love of variety and targeted advertising literatures, which have largely been separate, taking the other as given. Further, the firm side dynamics have been largely theoretical, and this paper seeks to provide empirical understanding of the effects of strategic advertising in a dynamic demand setting. Finally, the paper contributes to the broader literature on dynamic discrete choice models, by developing a model that incorporates both forward-looking and myopic consumers, and by providing an estimation strategy that can be applied to other settings.

2 Related Literature

2.1 Discrete Choice Demand Estimation

2.2 Variety-Seeking and Habit Formation

2.3 Advertising and Dynamic Demand

3 Model

3.1 Environment

There is a continuum of consumers $i \in \mathcal{I}$ who shop for $t \in \mathcal{T}$ periods. Each period t , they can make one purchase from a market. The market is made up of a continuum of firms $f \in \mathcal{F}$ in which one firm f^* acts as both the market and a producer. f^* will be the primary player, strategically advertising to consumer i based on inferences about their type. f^* will see all other $f \in \mathcal{F}$ as a competitor, denoted $\bar{f}$. The consumer is only concerned with maximizing their utility, while the firm wants to maximize profits.

3.2 Consumer's Problem

The consumer's problem is the following. In each period t , the consumer i chooses a product j from the menu of available products J_t . The consumer's utility from choosing product j at time t is given by:

$$U_{ijt} = \beta_i X_{jt} + \gamma_i \log(1 + (X_{jt} - \theta_{it})^2) + \alpha_i p_{jt}(1 - a_{it}) + \xi_j + \varepsilon_{ijt}$$

subject to

$$\theta_{it} = \{f(X_{js})\}_{s < t}$$

$$\gamma_i \sim \mathcal{N}(0, 1)$$

$$\varepsilon_{ijt} \sim \text{T1EV}(0, 1)$$

where β_i is a consumer's utility from characteristic X_{jt} , γ_i is their preference for variety, θ_{it} is a function of the consumers' past purchases. This could take the form of the mean characteristic of past purchases, the median characteristic, the norm, etc. α_i is a heterogeneous price sensitivity to p_{jt} , which is affected by some advertising a_{it} in the form of coupons. ξ_j is unobserved quality in product j , and ε_{ijt} is the Gumbel shock which allows for a closed form logit.

The choice probability of consumer i choosing product j in time t is as follows:

$$P_{ijt} = \frac{\exp U_{ijt}}{\sum_k \exp U_{ikt}}$$

I am currently working under a specification in which the consumer is a static agent, but hope to expand to a forward-looking learning agent after the simple case is handled.

3.3 Firm's Problem

The firm f^* chooses advertising a_{it} to maximize profits. The firm's problem is as follows:

$$\begin{aligned} V(a_{it}|\lambda_t) &= \max_{a_{it}} \{ \pi_{jt} + \mu \mathbb{E}[V(a_{it+1}|\lambda_{t+1})] \} \\ \pi_{jt} &= s(p_{jt}, a_{it})(p_{jt} - mc_{jt}) - c(a_{it}) \\ \lambda_t &\sim \mathcal{N}(\frac{1}{N} \sum_i \theta_{it}, \sigma_\lambda^2) \end{aligned}$$

Here, π_{jt} is the profit from advertising a_{it} , which depends on the market share $s(p_{jt}, a_{it})$, the price-cost margin $(p_{jt} - mc_{jt})$, and the cost of advertising $c(a_{it})$. The firm also has a discount factor μ for future profits, and faces uncertainty about the state variable λ_t , which is a function of the mean consumer history.

Other firms $\bar{f}$ are not strategic and do not advertise, but they do compete in the market, affecting the consumer's choice probabilities and thus the profits of f^* . Their problem is as follows:

$$\pi_{jt} = s(p_{jt})(p_{jt} - mc_{jt})$$

Most of the strategic dynamics will come from f^* , but the presence of $\bar{f}$ is important for identification and estimation.

3.4 Equilibrium

In equilibrium, consumers maximize utility via their consumption choices, and the firm maximized profits via advertising choices. The equilibrium concept is a Markov Perfect Equilibrium, in which the firm's advertising strategy is a function of the state variable λ_t , and the consumers' choice probabilities are consistent with their utility maximization given

the advertising strategy. The equilibrium conditions can be expressed as follows:

$$P_{ijt}^* = \frac{\exp U_{ijt}}{\sum_k \exp U_{ikt}}$$

$$a_{it}^* = \arg \max_{a_{it}} \{\pi_{jt} + \mu \mathbb{E}[V(a_{it+1} | \lambda_{t+1})]\}$$

where P_{ijt}^* is the choice probability of consumer i choosing the optimal product j at time t , and a_{it}^* is the optimal advertising strategy for firm f^* given the state variable λ_t .

4 Estimation

4.1 Data

To estimate the model, I am in the process of acquiring Nielsen's homescan and marketscan data. This data combination will provide detailed information on consumer purchases, such as products bought, prices paid, and quantities purchased. Further, I am able to see deals/coupons which are used as well as any advertising in the form of feature displays. The data has a panel structure, which allows me to implement these dynamics. The data also includes information on product characteristics, which will be important for estimating the utility function.

4.2 Identification of Random Coefficients

I plan to identify the random coefficients β_i , γ_i , and α_i using variation in product characteristics, prices, and advertising across consumers and over time. The presence of the strategic firm f^* and its advertising will provide additional variation that can help identify the effects of advertising on consumer preferences and demand. I will also leverage the panel structure of the data to control for unobserved heterogeneity across consumers.

4.3 GMM Objective and Moment Conditions

I will use a GMM estimation strategy to estimate the parameters of the model. The moment conditions will be derived from the first-order conditions of the consumer's utility maximization problem and the firm's profit maximization problem. Specifically, I will use the following

moment conditions:

$$\begin{aligned}\mathbb{E}[P_{ijt} - \frac{\exp U_{ijt}}{\sum_k \exp U_{ikt}}] &= 0 \\ \mathbb{E}[a_{it} - \arg \max_{a_{it}} \{\pi_{jt} + \mu \mathbb{E}[V(a_{it+1} | \lambda_{t+1})]\}] &= 0\end{aligned}$$

These moment conditions ensure that the estimated parameters are consistent with the observed choice probabilities and advertising strategies in the data.

4.4 CCP Approach

To solve the dynamic programming problem of the firm, I will use a Conditional Choice Probability (CCP) approach. This approach allows me to estimate the value function and the optimal advertising strategy without having to solve the full dynamic programming problem. Instead, I will estimate the CCPs directly from the data, and then use these estimates to recover the value function and the optimal advertising strategy. This approach is computationally efficient and can handle the high-dimensional state space that arises from the consumer's past consumption history and the firm's strategic advertising decisions.

5 Monte Carlo Simulation

5.1 Design

5.2 Fixed Mean Specification

5.3 Prior Mean Specification

5.4 Markovian Advertising Specification

6 Counterfactuals

6.1 Counterfactual 1

In my first idea, I want to evaluate the effects of a restriction on the strategic firm's ability to target based on competitors' purchase information. This could be motivated by a policy that restricts the use of data to consolidate market power. In this counterfactual, I will simulate the model under the assumption that strategic firm f^* now maximizes like the non-strategic firms $\bar{f}$, and does not have access to consumer history or preferences. This will allow me to

evaluate the effects of strategic advertising on consumer welfare and market outcomes, and to understand the implications of such a policy.

6.2 Counterfactual 2

Here, I am interested in evaluating the effects of perfect first-degree price discrimination by the strategic firm. This could be motivated by a scenario in which the strategic firm has access to perfect information about consumer preferences and is able to set personalized prices for each consumer. In this counterfactual, I will simulate the model under the assumption that strategic firm f^* can perfectly price discriminate, and evaluate the effects on consumer welfare and market outcomes.

7 Conclusion

In closing, this paper develops a dynamic discrete choice model of consumer demand that incorporates a love of variety and strategic advertising by firms. The model is estimated using scanner data, and counterfactual analyses are conducted to evaluate the effects of different market structures on consumer welfare and market outcomes. The paper contributes to the literature on advertising and consumer behavior, by providing empirical evidence on the effects of strategic advertising in a dynamic demand setting. The findings of this paper have important implications for policymakers and firms, as they highlight the potential benefits and drawbacks of targeted advertising in markets when accounting for consumers' preference for variety. Future research could extend this model to incorporate additional features, such as consumer learning, network effects, or multi-product firms, to further understand the dynamics of consumer demand and firm behavior in these markets.

A Proofs

A.1 Proof of Proposition ??

B Additional Tables and Figures

C Simulation Details